Rock, Paper, Scissors

Presented by GWC

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What you will Learn:

Random module
If, elif, else conditions
Quit function
Lists
Increment
Index

What Will the Program do?

- Store choices
- Take users input and computer input
- Decide winner

Let's get started!

Open replit and make a new file. Call it Rock Paper Scissors

Now in the first line type in the Import random module

main.py +

```
1  import random
2
```

Make a variable called computer_points and user_points and set it equal to zero. This will keep track of the points

main.py

```
1  import random
2
3  user_points = 0
4  computer_points = 0
5
```

Make a list called choices

```
user_points = 0
computer_points = 0

Choices = ["rock", "paper", "scissors"]
```

A list can store booleans, integers, strings and even other lists which are all called elements!

```
user_points = 0
computer_points = 0

Choices = ["rock", "paper", "scissors"]
```

Each elements or in this case string is represented by a numeric index that starts from 0

```
user_points = 0
computer_points = 0

Choices = ["rock", "paper", "scissors"]
```

```
Choices = ["rock", "paper", "scissors"]
```

0 1 2

On the next line add a while loop and the True boolean so that you can play rock paper scissors for as long as you want

```
while True:
```

Now below the while statement add the user input.

```
while True:
    user = input("Type rock/paper/scissor or type in q to quit: ")
```

After adding the user input add an if statement below. This will let us quit or exit the program.

```
while True:
    user = input("Type rock/paper/scissor or type in q to quit: ").lower()

    if user == "q":
        quit()
```

Add another if condition which will check if the user has typed in something invalid

```
if user == "q":
    quit()

if user_input not in choices:
```

If they have not typed in anything valid then continue will ask them the question again and again till the user types in something from the choice list

```
if user not in choices:
    continue
```

We have added the user input, so now we will add the computer input.

Whatever the computer picks should be random

```
if user_input not in choices:
    countinue

random_number = random.randint(0, 2)
```

To allow the computer to pick these choices, we will assign another variable so it can access the options.

```
random_number = random.randint(0, 2)
computer = choices[random_number]
```

To display what the computer chose, add a print statement.

```
random_number = random.randint(0, 2)
computer = choices[random_number]
print("Computer picked: " + computer)
```

Deciding winner
Using if condition

```
if user == "rock" and computer == "scissors":
    print("You won!")
    user_points += 1
```

Deciding winner continue...
Using elif condition

```
elif user == "paper" and computer == "rock"
    print("You won!")
    user_points += 1
```

Deciding winner continue...
Using elif condition

```
elif user == "scissors" and computer == "paper":
    print("You won!")
    user_points += 1
```

Deciding winner continue...
Using else condition

```
else:
    print("You lost!")
    computer_points += 1
```

When you run it, it should work.
But there is one setback in this program.

The problem is ---

if there is a tie the computer will win.

Example:
User and computer both choose rock

Finished code:

```python
import random
user_points = 0
computer_points = 0
choices = ["rock", "paper", "scissors"]
while True:
    user=input("Type rock/paper/scissors or type q to quit")
    if user == "q":
        quit()
    if user not in choices:
        continue
    random_number = random.randint(0, 2)
    computer = choices[random_number]
    print("Computer picked: " + computer)
    if user == "rock" and computer == "scissors":
        print("You won")
        user_points += 1
    elif user == "paper" and computer == "rock":
        print("You won")
        user_points += 1
    elif user == "scissors" and computer == "paper":
        print("You won")
        user_points += 1
    elif user == "rock" and computer == "rock":
        print("You tied")
        user_points += 0
        computer_points += 0
    elif user == "paper" and computer == "paper":
        print("You tied")
        user_points += 0
        computer_points += 0
    elif user == "scissors" and computer == "scissors":
        print("You tied")
        user_points += 0
        computer_points += 0
    else:
        print("You lost!")
        computer_points += 1
```

Thank you!